Supplemental material

Sex as a prognostic factor for mortality in critically ill adults with sepsis: a systematic review and meta-analysis

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Supplemental Table 1. Differences between the protocol and the review

Modified element	Explanation
Wording primary outcomes	We modify the wording for primary outcomes for clarity purposes, following the suggestion of peer reviewers. "All-cause hospital mortality" and "28-day all-cause mortality", instead of All-cause mortality (the longest follow-up provided by study authors)" and 28-day all-cause hospital mortality", respectively.
All-cause ICU mortality	We added all-cause ICU mortality as secondary outcome. We considered all-cause ICU mortality as a relevant outcome and non-subsidiary of pooling with hospital mortality outcomes.
Subgroup analyses	We were not able to undertake subgroup analyses comparing cohort versus case-control studies because there were insufficient studies.
Sensitivity analyses	We added sensitivity analysis after excluding the unique data from conference abstracts. We also carried out sensitivity analyses by pooling crude estimates. We were not able to perform the following sensitivity analyses specified in the protocol as no comparisons met the predefined criteria: - Excluding only studies with a high risk of bias in one QUIPS key domain. - Excluding studies that provided an adjusted estimated but did not adjusted for all our core set of additional prognostic factors.

Supplemental Table 2. Assessment of the use of terms sex and gender in the included studies

Adequate (any of the following):	Inadequate (any of following):
- Sex for biological characteristics. - Gender for socially constructed roles, behaviours, and identities. - Females or males for sex. - Women or men for gender.	- Gender for biological characteristics. - Sex for socially constructed roles, behaviours, and identities. - Females or males for gender. - Women or men for sex.

Supplemental Table 3. Process of defining the core set of adjustment factors

Step	Method	Potential additional prognostic factors identified
1. Preliminary searches to identify potential prognostic factors on mortality in patients with sepsis	1. PubMed search: (sepsis[Title]) AND "prognostic factor"[Title] 2. Embase: 'prognostic factor':ti AND 'sepsis':ti 3. Search in Uptodate 4. Initial discussion with review team members	1. Hypertriglyceridemia 2. Positive fluid balance 3. Red cell distribution width 4. Duration of SIRS before organ failure 5. Heart-type fatty acid-binding protein 6. D-dimer 7. Low serum level of high-density lipoprotein colesterol 8. Serum N-terminal pro-brain natriuretic peptide level 9. Immunosuppression 10. Cancer 11. Liver diseases 12. Alcohol dependence 13. Non-urinary source of infection 14. Inappropriate or late antibiotic coverage
2. Identify prognostic models for mortality in patients with sepsis	We considered factors included in the SOFA prognostic model	1. PaO2 2. FiO2 3. On mechanical ventilation 4. Platelets, $\times 10^3/\mu L$ 5. Glasgow Coma Scale 6. Bilirubin, mg/dL (µmol/L) 7. Mean arterial pressure OR administration of vasoactive agents required 8. Creatinine, mg/dL (µmol/L) (or urine output
3. Final list of key additional prognostic factors	We defined the final list of core set of adjustment factors by consensus	1. Age 2. Severity score at baseline (SOFA, SAPS II, APACHE II score) 3. Comorbidities: immunosuppression, pulmonary diseases, cancer, liver diseases, alcohol dependence 4. Non-urinary source of infection 5. Inappropriate or late antibiotic coverage

Supplemental Table 4. Search strategy

Full search string for MEDLINE Ovid (consulted 17th July 2020)
1. exp Sepsis/
2. exp Shock, Septic/
3. (septic* or sepsis* or SIRS).ti,ab.
4. "septic shock".ti,ab.
5. "endotoxic shock".ti,ab.
6. "toxic shock".ti,ab.
7. "severe sepsis".ti,ab.
8. "blood stream infection".ti,ab.
9. (septic?emia or "systemic inflammatory response syndrome" or py?emia).ti,ab.
10. (multi?organ adj5 failure).ti,ab.
11. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10
12. exp Sex Factors/
13. exp Sex Characteristics/
14. exp Sex Distribution/
15. exp Sex/
16. exp Sex Ratio/
17. exp Women's Health/
18. exp Men's Health/
19. boy*.ti,ab.
20. female*.ti,ab.
21. gender.ti,ab.
22. girl*.ti,ab.
23. male*.ti,ab.
24. men.ti,ab.
25. sex.ti,ab.
26. women.ti,ab.
27. 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26
28. 11 and 27
29. exp Mortality/
30. mortality.ti,ab.
31. dead.ti,ab.
32. death*.ti,ab.
33. died.ti,ab.
34. fatality.ti,ab.
35. fatalities.ti,ab.
36. survivor.ti,ab.
37. survival.ti,ab.
38. 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37
39. 28 and 38
40. incidence.sh.
41. follow up studies.sh.
42. "prognos*".ab,ti.
43. "predict*".ab,ti.
44. "course*".ab,ti.
45. 40 or 41 or 42 or 43 or 44
46. 39 and 45
47. exp Animals/ not humans.sh.

48. 46 not 47	
Full search string for Embase Elsevier (consulted 17th July 2020)	
#1	'sepsis'/mj
#2	'septic shock'/mj
#3	septic*:ab,ti OR sepsis*:ab,ti OR sirs:ab,ti
#4	'septic shock':ab,ti
#5	'endotoxic shock':ab,ti
#6	'toxic shock':ab,ti
#7	'severe sepsis':ab,ti
#8	'blood stream infection':ab,ti
#9	septic?emia:ab,ti OR 'systemic inflammatory response syndrome':ab,ti OR py?emia:ab,ti
#10	multi$organ NEAR/5 failure
#11	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10
#12	'sex factor'/mj
#13	'sexual characteristics'/mj
#14	'sex ratio'/mj
#15	'sex'/mj
#16	'women`s health'/mj
#17	'men`s health'/mj
#18	boy*:ab,ti
#19	female*:ab,ti
#20	gender:ab,ti
#21	girl*:ab,ti
#22	male*:ab,ti
#23	men:ab,ti
#24	sex:ab,ti
#25	women:ab,ti
#26	#12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25
#27	#11 AND #26
#28	'mortality'/mj
#29	mortality:ab,ti
#30	dead:ab,ti
#31	death:ab,ti
#32	died:ab,ti
#33	'fatality':ab,ti
#34	fatalities:ab,ti
#35	survivor:ab,ti
#36	survival:ab,ti
#37	#28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36
#38	#27 AND #37
#39	'disease course'/mj
#40	risk:kw
#41	diagnos*:kw
#42	'follow-up':kw
#43	epidemiology:lnk
#44	outcome:ab,ti
#45	#39 OR #40 OR #41 OR #42 OR #43 OR #44
#46	#38 AND #45
#47	'animal'/exp
#48	'human'/exp

#49 #47 NOT #48
#50 #46 NOT #49 AND ([embase]/lim OR [pubmed-not-medline]/lim)
Full search string for Web of Science (consulted 17th July 2020)
1 TOPIC: (sepsis) OR TOPIC: ("septic shock") OR TOPIC: ("Systemic inflammatory response syndrome") OR TOPIC: ("multiple organ failure")
2 TITLE: ("septic shock") OR TITLE ("endotoxic shock") OR TITLE: ("toxic shock") OR TITLE: ("severe sepsis") OR TITLE: ("blood stream infection") OR TITLE: (septic?emia) OR TITLE: (py?emia) OR TITLE: (septic*) OR TITLE: (sepsis*) OR TITLE: (SIRS)
3 #2 OR #1
4 TOPIC: ("sex factors" OR "sex distribution" OR "Sex characteristics" OR "Sex ratio" OR sex OR "women's health" OR "men's health") OR TITLE: (boy* OR male* OR girl* OR female* OR gender OR women OR men OR sex)
5 #4 AND #3
6 TOPIC: (mortality) OR TITLE: (mortality OR death OR dead OR died OR fatality OR fatalities OR survivor OR survival)
7 #6 AND #5
8 TOPIC: (incidende OR "follow up studies") OR TITLE: (prognos* OR predict* OR course*)
9 #8 AND #7
Trials registries (consulted 12th December 2019)
- ClinicalTrials.gov www.clinicaltrials.gov
- World Health Organization International Clinical Trials Registry Platform apps.who.int/trialsearch/
Hand-searched conference proceedings
- Interscience Conference on Antimicrobial Agents and Chemotherapy (ICAAC); 50th edition 2010 to 59th edition 2019.
- European Congress of Clinical Microbiology and Infectious Diseases (ECCMID); 20th edition 2010 to 29th edition 2019.
- Society for Healthcare Epidemiology of America (SHEA): IDWeek 2012 to 2019 editions.
- International Conference on Prevention and Infection Control (ICPIC): 2011, 2013, 2015, 2017, 2019
- Society of Critical Care Medicine (SCCM): 39th edition 2010 to 48th edition 2019.
- International Symposium on Intensive Care and Emergency Medicine (ISICEM): 30th edition 2010 to 39th edition 2019.
- European Society of Intensive Care Medicine (ESICM): 23rd edition 2010 to 32nd edition 2019.

Supplemental Table 5. Guide to judge the certainty of evidence for prognostic factors GRADE

We initially assigned high certainty of the evidence for phase-2 confirmatory designs, i.e., studies that sought to test independent associations between the prognostic factor and outcomes	
We considered that the following factors may downgrade the certainty of evidence:	
Risk of bias	We rated as having: 1) serious limitations when most evidence was from studies at moderate or unclear risk of bias for most of the QUIPS domains; 2) very serious limitations when most evidence was from studies at high risk of bias for most of the QUIPS domains.
Inconsistency	We judged inconsistency relying on variability in point estimates using prediction intervals, extent of overlap of these intervals, and considering where point estimates lie in relation to clinical decision thresholds. We pre-specified subgroup analyses to explore differences across categories. In case of a single study within the existing body of evidence estimated the effect, we considered this criterion as "not applicable".
Indirectness	We downgraded the certainty of evidence whether participant population, prognostic factor, and/or outcomes fully represented no the review question. We judged indirectness for the prognostic factor based on characteristics of the primary independent variable, regardless of the adequacy of used terms, since we assessed insufficient details of sex and gender definitions provided or non-stated in the prognostic factor measurement QUIPS domain.
Imprecision	We judged imprecision considering: - Optimal information size - Compatibility of the 95% confidence interval of the absolute risk difference with our pre-defined clinical thresholds (minimal prognostic effects that were considered as clinically relevant for decision-making)
Publication bias	We planned to assess the presence of publication bias for each meta-analysis containing ≥10 studies by funnel plot representation and Peter's test at a 10% level.
We considered that the following factors may upgrade the certainty of evidence:	
Large effect estimate	We assessed size effect estimate considering: i) For meta-analysis: We considered upgrading the certainty of evidence for moderate or large pooled effects. Arbitrary thresholds define moderate odds ratio (1.5 ≤ OR ≤ 2), or large (OR > 2) ii) For narrative summary: We considered upgrading the certainty of evidence for moderate or large effects reported by most of the primary studies.
Dose response	We considered no dose response because of the feature of our prognostic factor of interest (dichotomous)

Abbreviations: OR: Odds ratio; QUIPS: Quality in prognosis studies.

Supplemental Table 6. Descriptive summary of included studies

	Adrie 2017	Caceres 2013	Dara 2012	Luethi 2020	Madsen 2014	Mahmood 2012	Nachtigall 2011
Methods							
Study design	Nested case-control	Cohort	Cohort	Post-hoc analysisis	Cohort	Cohort	Cohort
Database	OutcomeRea	IMPACT-HAP	CATSS	ARISE	SSC Database	APACHE IV	Not reported
Sample size calculation	Not reported	Not reported	Not reported	Not reported	Reported	Not reported	Not reported
Participants*							
Females; Males	631 (37); 1,061 (63)	145 (35); 271 (65)	3667 (42.3); 5003 (57.7)	562 (40.5); 825 (54)	365 (45); 449 (55)	13221 (47.3); 14714 (52.7)	130 (40); 197 (60)
Sociodemographics							
Age	69 (57-77); 65 (51-75)	62.4 (16.9); 55.7 (16.5)	62.8 (15.9) ; 62.3 (16.6)	62 (17.1); 63.5 (15.8)	66.2 (18); 66.3 (16.2)	Not reported §	68 (57-78); 64 (50-72)
Race	Not reported	Not reported		Not reported	284 (78.5); 370 (82.4)	Not reported §	Not reported
Caucasian							
African-American							
Latin							
Other/unknown							
Socioeconomic status	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported
Comorbidities						Not reported §	
Respiratory	123 (19.5); 266 (25)	37 (25.5); 54 (20.2)	Not reported	53 (8.3); 90 (9.5)	Not reported		20 (15.4); 27 (13.7)
Cardiac	71 (11.2); 123 (11.6)	32 (22.1); 58 (21.6)	(9.9) ; (11.2)	46 (7.2); 128 (13.5)	77 (21.1); 97 (21.6)		Not reported
Renal	19 (3); 29 (2.7)	31 (21.4); 45 (16.9)	Not reported	21 (3.3); 43 (4.5)	Not reported		24 (18.5); 37 (18.8)
Diabetes	Not reported	46 (31.7); 74 (27.6)	Not reported	27 (4.2); 58 (6.1)	Not reported		Not reported
Immunosuppression	119 (18.9); 207 (19.5)	60 (41.4); 101 (37.8)	(1.8) ; (3.3)	Not reported	Not reported		18 (13.8); 11 (5.6)
Liver disease	28 (4.4); 66 (6.2)	Not reported	(10) ; (16.9)	26 (4.1); 57 (6)	Not reported		8 (6.2); 17 (8.6)
Cancer	Not reported	Not reported	Not reported	87 (13.6); 161 (16.9)	Not reported		Not reported
Severity score							
APACHE II	19 (14-24); 19 (14-24)	22.1 (7.6); 19.9 (7.2)	25.9 (8.2) ; 25.5 (8.1)	48.1 (20.4); 50.2 (20.0) ‡	Not reported		Not reported
SAPS II	44 (33-58); 45 (34-60)	Not reported	Not reported	Not reported	Not reported		40 (29-53); 39 (28-51)
SOFA	6 (4-9); 6 (4-9)	Not reported	Not reported	3.7 (2.7); 4.2 (2.8)	6.2 (2.9); 7.2 (3.2)		5 (3-7); 6 (4-9)
Infection site							
Urinary source of infection	68 (10.8); 51 (4.8)	N/A	Not reported	138 (21.6); 170 (17.9)	Not reported	Not reported §	31 (23.8); 14 (7.1)
Prognosis factor							
Independent variable	Gender	Gender	Gender	Gender	Gender	Gender	Gender
Sex/ gender definition	Not reported	Not reported	Not reported	Reported	Not reported	Not reported	Not reported
Terms used	Gender, sex, female, male, woman/men, man/men	Gender, sex, female, male, woman/men, man/men	Gender, female, male	Gender, sex, female, male, woman/men, man/men	Gender, sex, female, male, woman/men, man/men	Gender, female, male, woman/men, man/men	Gender, sex, female, male, woman/men, man/men
Appropriateness of terms use	Inadequate	Inadequate	Inadequate	Unclear	Inadequate	Inadequate	Inadequate

Extracted outcomes							
Primary outcomes							
All-cause hospital mortality	Yes	Yes	Yes	No	Yes	No	No
28-day all-cause mortality	No	Yes	No	No	No	No	No
Secondary outcomes							
7-day all-cause hospital mortality	No	No	No	No	No	No	No
1-year all-cause mortality	No	No	No	No	No	No	No
All-cause ICU mortality	Yes	No	Yes	Yes	No	Yes	Yes
Follow-up	Not reported	Hospital discharge, death or 28 days after penumonia diagnosis, whichever ocurred first	Not reported	Not reported	Not reported	Not reported	Not reported
Identification							
Country	France	United States	Canada, United States, Saudi Arabia	Australia, New Zealand, Finland, Hong Kong, Ireland	United States	United States	Germany
Funding source	Educational grants from Aventis Pharma, France, and Wyeth; and public funds	Pfizer. University of Lousville Foundation responsible for project oversight	Unrestricted grants from Eli-Lilly, Pfizer, Bayer, Astellas, Merck, Mantioba Research Council, Health Sciences Centre Foundation, Innovations and Opportunities Foundation, Deacon Foundation	National Health and Medical Research Council	Alpert Medical School of Brown University	Not reported	Not reported
Conflict of interest	None	Declared	Declared	Declared	Not reported	None	Declared
Identifier or protocol	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported
Notes	Authors used conditional logistic regression with matching on age, death propensity score, and center. Email sent to study authors in May 2020; no reply received.	28-day mortality reported, authors were contacted for clarification in May 2020; no reply received.	Email sent to study authors in March 2020; no reply received	Baseline data available only for main cohort (N=1,591 participants). Email sent to study authors in May 2020; reply received but we were unable to get additional data.	Email sent to study authors in May 2020; no reply received.	Baseline data available only for main cohort (N=261,255 participants) Email sent to study authors in June 2020; no reply received.	Email sent to study authors in May 2020; no reply received.

Continued

	Pietropaoli 2010	Sakr 2013	Samuelsson 2015	Sunden-Cullberg 2020	van Vught 2017	Xu 2019
Methods						
Study design	Cohort	Cohort	Cohort	Cohort	Cohort	Cohort
Database	Cerner Project IMPACT	Piademont Intensive Care Unit Network	SIR	NQSR and SIR	MARS	MIMIC-III
Sample size calculation	Reported	Not reported	Not reported	Not reported	Not reported	Not reported
Participants*						
Females; Males	8,702 (46); 10,055 (54)	85 (27.9); 220 (72.1)	Not reported §	1,210 (44.5); 1,510 (55.5)	595 (38.8); 938 (61.2)	2,677 (43.6); 3,457 (56.4)
Sociodemographics						
Age	68 (54-75); 65 (52-76)	67.7 (14.3); 63.1 (15)	Not reported §	68 (56–77); 68(58–77)	59.4 (16.2); 60.8 (14.8)	65-89 (50.4); 65-89 (51.1)‖
Race		Not reported	Not reported §	Not reported		
Caucasian	6,439 (74); 7,541 (75)				510 (85.7); 839 (89.4)	1,915 (71.5); 2,597 (75.1)
African-American	1,218 (14); 1,207 (12)					369 (13.8);273 (7.9)
Latin	435 (5); 603 (6)					70 (2.6); 143 (4.1)
Other/unknown	610 (7); 704 (7)					238 (8.9); 325 (9.4)
Socioeconomic status	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported
Comorbidities			Not reported §	Not reported		Not reported
Respiratory	870 (10); 1005 (10)	3 (3.5); 18 (8.2)			72 (12.1); 138 (14.7)	
Cardiac	522 (6); 704 (7)	8 (9.4); 17 (7.7)			131 (22); 232 (24.7)	
Renal	522 (6); 603 (6)	16 (18.8); 40 (18.2)			86 (14.5); 131 (14)	
Diabetes	Not reported	18 (21.2); 34 (15.5)			124 (20.8); 183 (19.5)	
Immunosuppression	1,131 (13); 1,307 (13)	Not reported			Not reported	
Liver disease	261 (3); 402 (4)	Not reported			Not reported	
Cancer	1,218 (14); 1,709 (17)	4 (4.7); 6 (2.7)			136 (22.9); 245 (26.1)	
Severity score						
APACHE II	21 (15-27); 21 (15-27)	Not reported		Not reported	79 (62-99); 76 (58-98)†	Not reported
SAPS II	35 (15-64); 33 (14-64)	55 (18.8); 55.3(17.5)		64 (55–73); 65 (56–75)¶	Not reported	21.39 (5.73); 21.06 (5.6)
SOFA	Not reported	9.1 (3.3); 9.8 (3.7)		Not reported	7 (5-9); 7 (4-9)	6.97 (3.52); 7.29 (3.75)
Infection site						
Urinary source of infection	2,698 (31); 1,910 (19)	5 (5.9); 13 (5.9)	Not reported §	258 (21.3); 301(19.9)	Not reported	Not reported
Prognosis factor						
Independent variable	Gender	Gender	Gender/Sex	Sex	Gender	Sex
Sex/ gender definition	Reported	Not reported	Not reported	Reported	Not reported	Not reported
Terms used	Gender, sex, female, male, woman/men, man/men	Gender, sex, female, male, woman/men, man/men	Gender, sex, female, male, woman/men, man/men	Gender, sex, female, male, woman/men, man/men	Gender, sex, female, male, woman/men, man/men	Gender, sex, female, male, woman/men, man/men
Appropriateness of terms use	Inadequate	Inadequate	Inadequate	Inadequate	Inadequate	Inadequate
Extracted outcomes						
Primary outcomes						
All-cause hospital mortality	Yes	No	No	No	Yes	Yes

28-day all-cause mortality	No	Yes	Yes	Yes	Yes	Yes
Secondary outcomes						
7-day all-cause hospital mortality	No	No	No	No	No	No
1-year all-cause mortality	No	No	No	No	Yes	Yes
All-cause ICU mortality	Yes	Yes	No	No	Yes	No
Follow-up	Not stated	Death or ICU discharge	30 days	30 days	1 year	1 year
Identification						
Country	Brazil, Canada, US	Italy	Sweden	Sweden	Netherlands	United States
Funding source	National Heart, Lung and Blood Institute	Regione Piamonte, *progetti finalizzati di ricerca*	Regional Health Care Authorities in the Halland and Skåne regions of Sweden	Karolinska Institute, Swedish Government Funds for Clinical Research	Center for Translational Molecular Medicine, project MARS	Guangzhou Science and Technology Programs, the Guangdong Provincial Key Laboratory Construction Projection on Organ and Transplant Immunology, and the Guangdong Provincial International Cooperation Base of Science and Technology
Conflict of interest	None	None	None	None	Declared	None
Identifier or protocol	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported
Notes	Email sent to study authors in April 2020; no reply received.	ICU mortality mismatched published data, authors were contacted for clarification in April 2020; reply received. 28-day mortality reported, authors were contacted again for clarification in May 2020; no reply received.	30-day mortality reported, authors were contacted for clarification in June 2020; reply received (outcome included 30-day in- and out-hospital mortality). Sepsis subgroup comparison was adjusted at P<0.001.	30-day mortality reported, authors were contacted for clarification in June 2020; no reply received.	30-day mortality reported, authors were contacted for clarification in May 2020; no reply received.	Cox analyes reported as OR without additional clarification, and 30-day mortality reported, authors were contacted for clarification in July 2020; no reply received.

*Categorical variables expressed as numerical values and percentages, and continuous variables expressed as median and IQR, or mean and standard deviation as the study may be.

† APACHE IV

‡ APACHE III

§ Participant characteristics only available for whole ICU cohort

¶ SAPS III

|| Age reported by the study authors as percentage of participants in different age groups. Age expressed as age group (percentage).

Abbreviations: APACHE: Acute Physiology and Chronic Health Evaluation; ARISE: Australasian resuscitation in sepsis evaluation; CATSS: Cooperative antimicrobial therapy of septic shock; ICU: Intensive care unit; IMPACT: abbreviation not detailed; IMPACT- HAP: Improving medicine through pathway assessment of critical therapy in hospital-acquired pneumonia; F: Females; M: Males; MARS: Molecular diagnosis and risk stratification of sepsis; MIMIC: Medical information mart for intensive care III; N/A: Not applicable; NQSR: National quality sepsis registry; SAPS: Simplified Acute Physiology Score; SIR: Swedish intensive care registry; SOFA: Sequential Organ Failure Assessment score; SSC: Surviving sepsis campaign.

Supplemental Table 7. Sepsis definition provided by the study authors

Study	Sepsis-related term for defining health condition	Operational definition
Adrie 2007*	Sepsis severe	Severe sepsis was defined as infection with two or more criteria for systemic inflammatory response syndrome and at least one criterion for organ dysfunction
Caceres 2013	Severe infection, hospital-acquired pneumonia	Severe infection was defined as hospital-acquired pneumonia, including ventilator-associated pneumonia and health-care associated pneumonia
Dara 2012	Sepsis shock	Non-provided
Luethi 2010	Septic shock	Septic shock was defined as two or more criteria for systemic inflammatory response syndrome and refractory hypotension (systolic blood pressure of b90 mmHg or a mean arterial pressure of 65 mmHg after an intravenous fluid challenge), or hyperlactatemia (blood lactate level of ≥4.0 mmol/L), or both.
Madsen 2014	Severe sepsis and septic shock	Severe sepsis or septic shock as defined by Surviving Sepsis Campaign.
Mahmood 2012	Sepsis	Non-provided
Nachtigall 2011	Sepsis	Sepsis, severe sepsis, and septic shock was defined according to the national and international sepsis guidelines, requering two or more criteria for systemic inflammatory response syndrome associated with an infection
Pietropaoli 2010	Severe sepsis and septic shock	Severe sepsis was defined as development of at least one severe acute organ dysfunction within 3 days of a presumed infection.
Sakr 2013	Severe sepsis	Sepsis syndromes were diagnosed according to the criteria proposed by the American College of Chest Physicians/ Society of Critical Care Medicine Consensus Conference [Severe sepsis: sepsis associated with organ dysfunction, hypoperfusion, or hypotension]
Samuelsson 2015	Sepsis	Non-provided
Sunden-Cullberg 2020	Severe sepsis and septic shock	Severe sepsis and septic shock were diagnosed using a modified version of the 1992 sepsis definition, in practice accepting a diagnosis of severe sepsis on the basis of infection plus organ dysfunction
van Vught 2017†	Sepsis	Sepsis was defined as an infection diagnosed with a "probable" or "definite" likelihood, plus at least one additional variable as described in the 2001 International Sepsis Definitions. Shock was defined by the use of vasopressors.
Xu 2019‡	Sepsis, severe sepsis and shock septic	Non-provided

Supplemental Table 8. Prognostic factors in adjusted models for mortality in included studies

Study	Prognostic factors included in adjusted analyses
Adrie 2007*	Chronic respiratory failure; metastatic cancer; immunocompromised status; emergency surgery; acute respiratory failure and shock at hospital admission; urinary tract infection as a cause of sepsis; type of microorganism (*E coli, S pneumoniae*, and *Enterobacter* species)
Caceres 2013	Age; APACHE II; HCAP; white race; history of cardiac/renal/vascular/diabetes/respiratory disease; severe sepsis; hospital LOS; ICU LOS; MV after diagnosis of MRSA; CPIS at baseline
Dara 2012	APACHE II; age; site of infection; source of admission ; inappropriate antibiotics; other variables related to organ dysfunction
Luethi 2010	Illness severity (APACHE III score); pre-existing comorbidities (Charlson comorbidity index); cardiac arrhythmia; intravenous resuscitation fluid (per kilogram) administered before ICU admission
Madsen 2014	Age; race; SOFA; CHF; coagulopathy
Mahmood 2012	Acute physiology score; age; ethnicity; pre-ICU length of stay; pre-ICU location and hospital teaching status
Nachtigall 2011	Age; TISS-28 on admission (nursing workload); occurrence of pneumonia; septic shock; fungi detected; septic shock
Pietropaoli 2010	Age; dependent functional status at admission; African-American race; type of admittance; medical versus surgical patient; type of insurance; CPR within 24h of admission; comorbidities (chronic liver disease, active cancer within 5 years, chronic cardiovascular disease, chronic respiratory disease, immunocompromised status); illness severity (neurological dysfunction, cardiovascular dysfunction, elevated serum lactate, acute renal failure, hepatic dysfunction, hematologic dysfunction; SAPS II score); source of infection; processes of care; hospital characteristics
Sakr 2013	Age; comorbidities (renal failure with dialysis, chronic obstructive pulmonary disease); SAPS II; type of admission (elective surgery, emergency surgery, medical admission); initial SOFA sub-scores; referring facility; source of infection (abdominal)
Samuelsson 2015	Age; comorbidity (scored as in the Simplified Acute Physiology III); hospital LOS in days; location prior to ICU admission; therapy prior to ICU admission; reason for ICU admission; surgical status; presence of nosocomial or lower-airway infection; physiologic derangement (scored as in the Simplified Acute Physiology III); hospital characteristics
Sunden-Cullberg 2020	Temperature-adjusted SAPS3; body temperature; incorrect antibiotics; treatment limitations
van Vught 2017†	Age; body mass index; comorbidity; source of infection; acute physiology score
Xu 2019‡	Age; race; first ICU service; marital status; insurance; admission location; SAPS; SOFA

* Adrie 2007 reported adjusted analyses using a conditional logistic regression after matching on age, death propensity score, and centre.

† van Vught 2017 reported adjusted analyses only for 90-day mortality.

‡ Xu 2019 reported adjusted analyses using a Cox proportional hazard regression model.

Abbreviations: APACHE: Acute Physiology and Chronic Health Evaluation; CHF: Congestive heart failure; CPIS: Clinical Pulmonary Infection Score; CPR: Cardiopulmonary resuscitation; HCAP: Health care-associated pneumonia; ICU: Intensive care unit; MRSA: methicillin-resistant *Staphylococcus aereus;* MV: Mechanical ventilation; LOS: Length of stay; SAPS: Simplified Acute Physiology Score; SOFA: Sequential Organ Failure Assessment score; TISS-28: Therapeutic Intervention Scoring System-28.

Supplemental Table 9. Summary outcome estimates for each included study

Study	Unadjusted OR, 95%CI*				Adjusted OR, 95%CI*			
	Hospital mortality	28-day mortality	1-year mortality	ICU mortality	Hospital mortality	28-day mortality	1-year mortality	ICU mortality
Adrie 2007	0.88 (0.71-1.10)	N/A	N/A	0.87 (0.69-1.09)	0.75 (0.57-0.97)	N/A	N/A	0.75 (0.58-0.98)
Caceres 2013	1.35 (0.81-2.26)	1.35 (0.81-2.26)	N/A	N/A	0.99 (0.52-1.93)	0.99 (0.52-1.93)	N/A	N/A
Dara 2012	0.95 (0.87-1.04)	N/A	N/A	N/A	1.07 (0.96-1.19)	N/A	N/A	N/A
Luethi 2010	N/A	N/A	N/A	1.14 (0.82-1.58)	N/A	N/A	N/A	<50y: 1.18 (0.47-2.86) >50y: 1.33 (0.90-1.96)
Madsen 2014	1.10 (0.80-1.52)	N/A	N/A	N/A	"Multivariable analysis…Gender was not associated with in-hospital survival"	N/A	N/A	N/A
Mahmood 2012	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1.07 (0.99-1.16)
Nachtigall 2011	N/A	N/A	N/A	1.89 (1.06-3.36)	N/A	N/A	N/A	1.91 (1.00-3.64)
Pietropaoli 2010	1.09 (1.02-1.16)	N/A	N/A	1.09 (1.02-1.17)	1.11 (1.04-1.19)	N/A	N/A	N/A
Sakr 2013	N/A	"Kaplan-Meier analysis showed reduced 28-day survival in female compared with male patients"	N/A	2.01 (1.20-3.37)	N/A	N/A	N/A	2.23 (1.17-4.24)

Samuelsson 2015	N/A	N/A	N/A	N/A	N/A	1.17 (1.06-1.29)†	N/A	N/A
Sunden-Cullberg 2020	N/A	1.11 (0.91-1.36)	N/A	N/A	N/A	1.28 (1.00-1.64)	N/A	N/A
van Vught 2017‡	1.02 (0.81-1.27)	1.13 (0.90-1.43)	0.92 (0.74-1.13)	1.14 (0.89-1.45)	N/A	N/A	N/A	N/A
Xu 2019	0.89 (0.80-0.99)	0.91 (0.82-1.01)	0.84 (0.76-0.93)	N/A	N/A	N/A	0.83 (0.68-0.98)§	N/A

***** Prognostic effect reported as OR (95% CI).

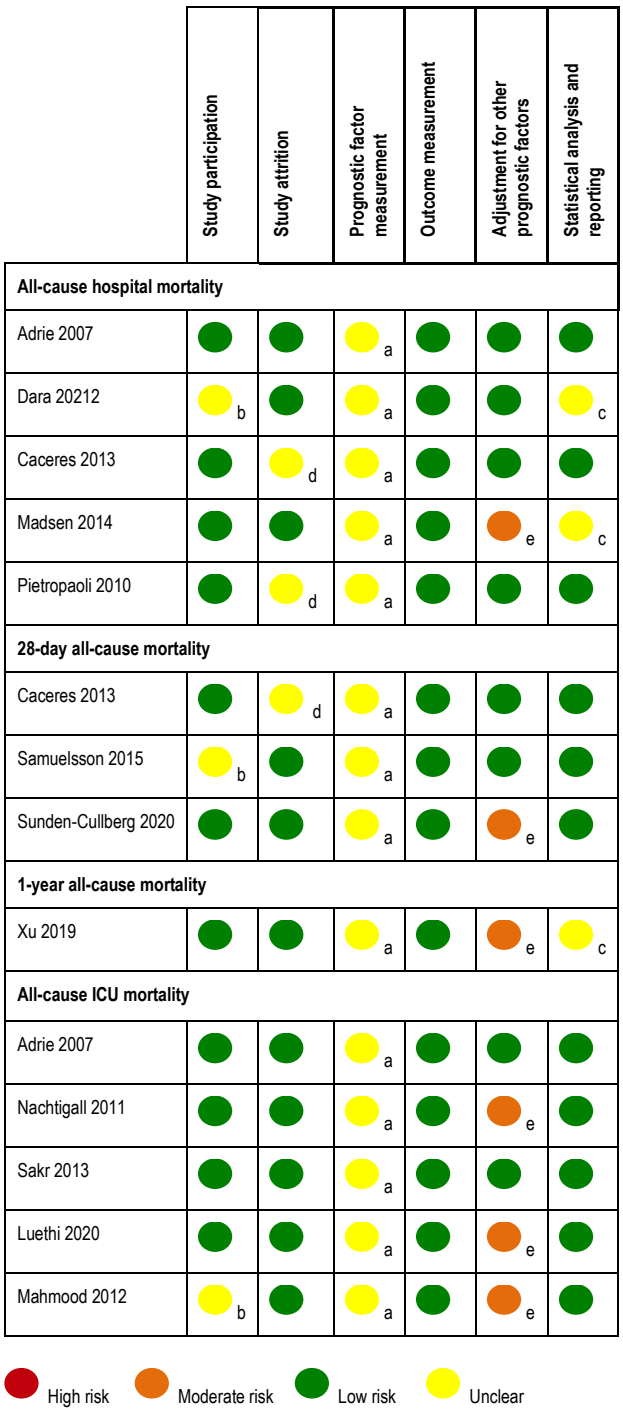
† Prognostic effect reported by the study authors as OR (99% CI), 1.17 (1.03- 1.33). We transformed it into OR (95% CI).

‡ van Vught 2017 reported adjusted analyses only for 90-day mortality.

§ Xu 2019 reported adjusted analyses using a Cox proportional hazard regression model as OR (95% CI), 1.08 (1.01-1.17), without additional clarifications. After contacting the study authors and no reply received, we assumed that they reported Cox analyses as hazard ratios (HR). We transformed HR into OR (95% CI)

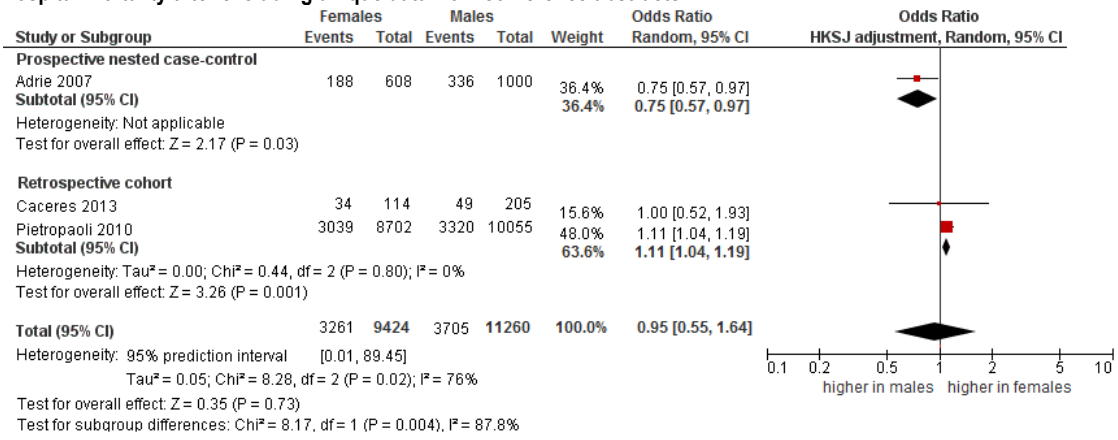
Abbreviations: CI: Confidence interval; N/A: Not available; OR: Odds ratio; Y: Years old.

Supplemental Figure 1. QUIPS Risk of bias domain summary by outcome

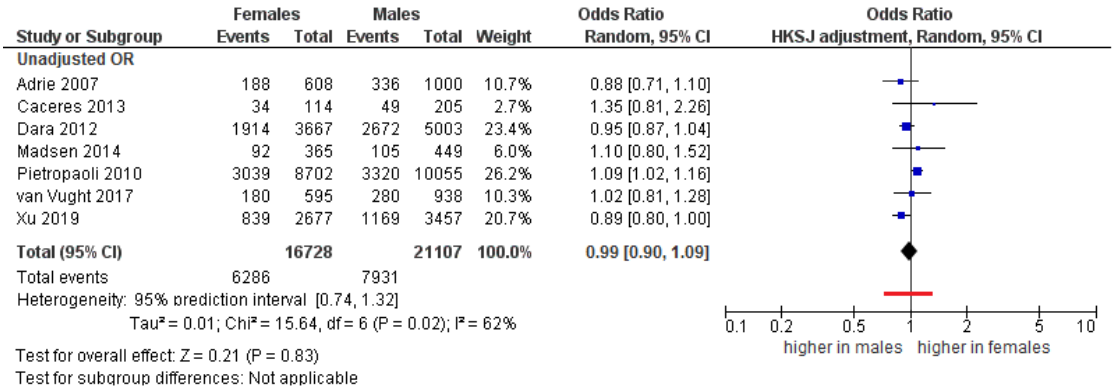


Explanations:
a. Unclear or not stated a definition of sex or gender.
b. Insufficient data on baseline description for sepsis subgroup.
c. Insufficient presentation of data to assess the adequacy of the analytic strategy.
d. Inadequate description of dropouts to judge the risk of important differences between participants analysed and those who were not.
e. Minimal adjustment for covariates as defined in our review core set of adjustment factors.

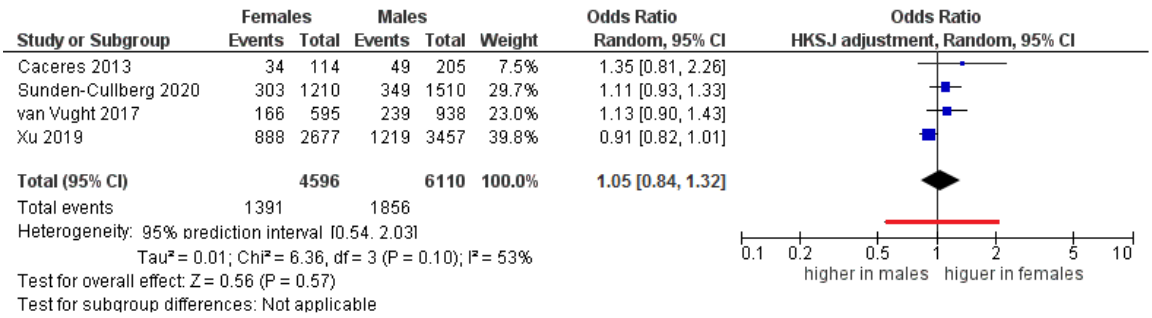
Supplemental Figure 2. Sensitivity analysis of adjusted analyses for association between sex and all-cause hospital mortality after excluding unique data from conference abstracts



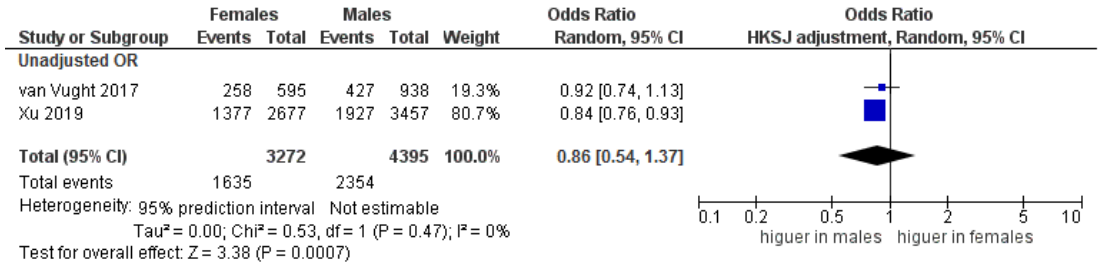
Supplemental Figure 3. Forest plot of unadjusted analyses for association between sex and all-cause hospital mortality



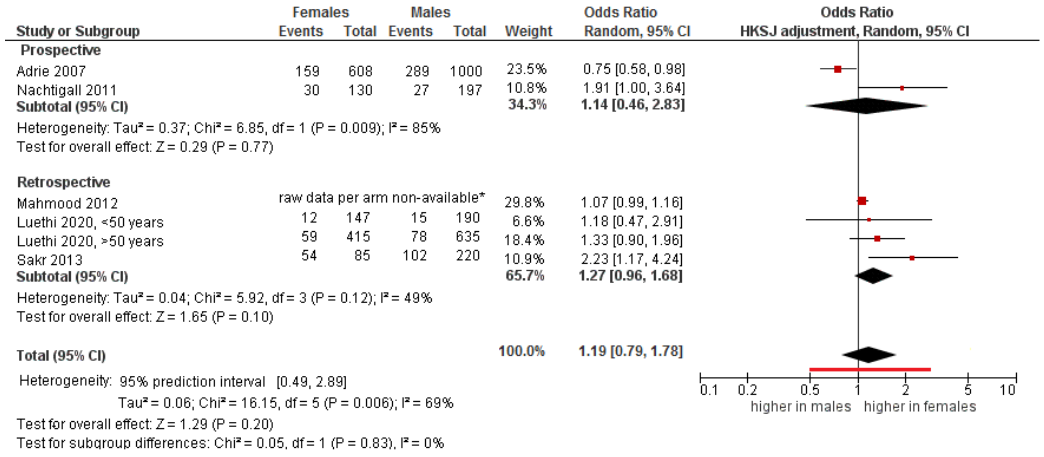
Supplemental Figure 4. Forest plot of unadjusted analyses for association between sex and 28-day all-cause mortality



Supplemental Figure 5. Forest plot of unadjusted analyses for association between sex and 1-year all-cause mortality

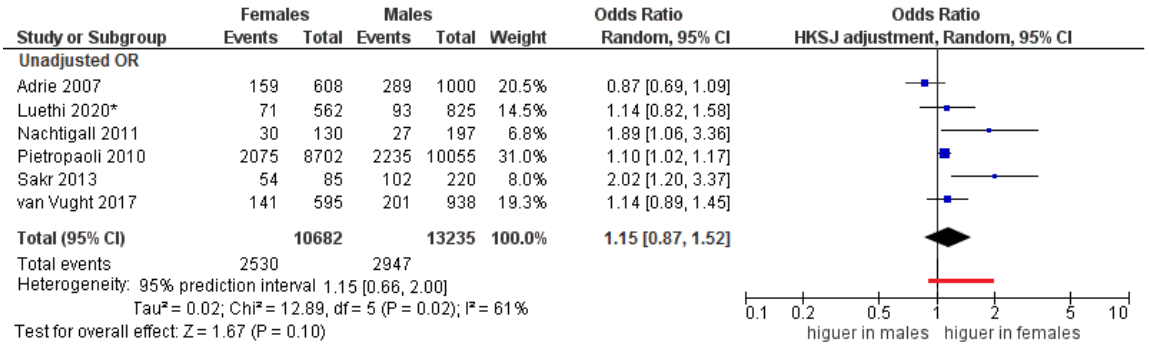


Supplemental Figure 6. Forest plot of adjusted analyses for association between sex and all-cause ICU mortality



* only provided the adjusted estimate

Supplemental Figure 7. Forest plot of unadjusted analyses for association between sex and all-cause ICU mortality



* Luethi 2020 reported an overall unadjusted odds ratio.